Here’s a step-by-step guide to create a simple web application that demonstrates prefix grouping with route namespaces in Laravel 12.

4.11.4.1 Step 1: Create and Open Laravel Project

We will create a new Laravel project named `lab-prefix` using Composer. Open your terminal and run:

```
composer create-project laravel/laravel:^12.0.3 lab-prefix
cd lab-prefix
code .
```

You should see the Laravel project structure in Visual Studio Code.

4.11.4.2 Step 2: Create Namespaced Controllers

Create a folder and two controllers using Artisan:

```
php artisan make:controller Admin/DashboardController
php artisan make:controller Admin/UserController
```

This will create two controllers in the `app/Http/Controllers/Admin/` directory:

```
app/Http/Controllers/Admin/DashboardController.php
app/Http/Controllers/Admin/UserController.php
```

4.11.4.3 Step 3: Define Route Groups with Prefix and Controller Namespace

We’ll define a route group with a prefix of `admin` and map it to the `Admin` namespace. This allows us to keep our routes organized and easy to manage.

Edit `routes/web.php`:

```
use App\Http\Controllers\Admin\DashboardController;
use App\Http\Controllers\Admin\UserController;

Route::prefix('admin')->group(function () {
    Route::get('/dashboard', [DashboardController::class, 'index'])->name('admin.dashboard
    Route::get('/users', [UserController::class, 'index'])->name('admin.users');
    Route::get('/users/{id}', [UserController::class, 'show'])->name('admin.users.show');
});
```

Save the file.